

# Yaoteng Tan

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## RESEARCH INTERESTS

Trustworthy AI, foundation models, machine unlearning, adversarial robustness, vision-language models, text-to-image generation, AI safety, and optimization for machine learning.

## EDUCATION

**University of California, Riverside**

Sep 2022 – Jun 2027 expected

*Ph.D. Candidate in Electrical and Computer Engineering*

*Riverside, CA*

Advisor: Prof. [M. Salman Asif](#)

**Huazhong University of Science and Technology**

Sep 2018 – Jun 2022

*B.S. in Electrical Engineering, Graduated with Honors*

*Wuhan, China*

## PUBLICATIONS

1. **Y. Tan\***, Z. Cai\*, M. S. Asif. *Targeted Unlearning with Single Layer Unlearning Gradient*. Proceedings of the **International Conference on Machine Learning (ICML)**, 2025. [\[arXiv\]](#) [\[project page\]](#)
2. **Y. Tan\***, Z. Cai\*, M. S. Asif. *Ensemble-based Blackbox Attacks on Dense Prediction*. Proceedings of the **IEEE Conference on Computer Vision and Pattern Recognition (CVPR)**, 2023. [\[arXiv\]](#) [\[code\]](#)
3. **Y. Tan**, Z. Cai, M. S. Asif. *Modular Energy Steering for Safe Text-to-Image Generation with Foundation Models*. Under review at ECCV 2026. [\[arXiv\]](#)
4. **Y. Tan**, Z. Cai, M. S. Asif. *Transform-dependent Adversarial Attacks*. Under review at TMLR [\[arXiv\]](#)

\* Equal contribution.

## RESEARCH EXPERIENCE

**Graduate Research Assistant**

Oct 2022 – Present

*University of California, Riverside*

*Riverside, CA*

- Conduct research on trustworthy AI, with emphasis on machine unlearning, adversarial robustness, and safety mechanisms for foundation models.
- Develop efficient algorithms for vision-language models and text-to-image generative models, targeting both empirical performance and mechanistic understanding.
- Design and evaluate methods across modern machine learning benchmarks, including adversarial attack, unlearning, and generative safety settings.

**Research Engineer Intern**

Jun 2026 – Sep 2026 expected

*Center for AI Safety*

*San Francisco, CA*

Mentor: Dr. [Steven Basart](#)

- Contributing to AI safety projects that are endeavour for societal impact, open to public, targeting publication in major machine learning venues.

**Machine Learning Engineer Intern**

Jul 2021 – Aug 2021

*CloudMinds Technology Inc.*

*Beijing, China*

- Developed a real-time robotics vision module using object detection and deployed it in an Unreal Engine simulation environment for testing.

## SELECTED RESEARCH PROJECTS

**Modular Energy Steering for Safe Text-to-Image Generation**

Aug 2025 – Mar 2026

[arXiv:2604.02265](#) | *Keywords: AI safety, scalable oversight, Text-to-image models*

- Proposed a training-free, and dataset-free inference-time safety steering framework that leverages

pretrained foundation models as oversight signals for text-to-image generation.

- Demonstrated compatibility with diffusion and flow-matching models, including scalable steering for multiple target concepts.
- Reduced average NSFW attack success rate from 82.7% to 8.0% across four red-teaming benchmarks on SD-v1.4, while preserving image quality on general concepts.

### **Targeted Unlearning with Single Layer Unlearning Gradient**

Jun 2024 – Jul 2025

[ICML 2025](#) | *Keywords: machine unlearning, post-training adaptation, vision-language models*

- Proposed SLUG, a one-step gradient-based targeted unlearning for multimodal foundation models.
- Achieved 20–50× higher efficiency and improved retention-forgetting trade-off over existing baselines.
- First showed that targeted unlearning can be performed as a layer-wise modular update, enabling localized concept removal while preserving retained knowledge.

### **Discovery of Sparsity-Constrained Optimization Algorithms**

Sep 2024 – Jul 2025

[GitHub repository](#) | *Keywords: optimization, meta-learning, AI for science*

- Developed a meta-optimization framework for discovering adaptive solvers for sparsity-constrained inverse problems.
- Connected classical optimization structure with data-driven algorithm design for sparse recovery.

### **Transform-dependent Adversarial Attacks**

Sep 2022 – Jun 2023

[arXiv:2406.08443](#) | *Keywords: adversarial attacks, physical robustness, imaging pipeline*

- Introduced transform-dependent adversarial attacks that reveal vulnerabilities beyond conventional static perturbation settings.
- Built a camera pipeline that enables optimization through hardware-related image transformations.

### **Ensemble-based Blackbox Attacks on Dense Prediction**

Sep 2022 – Jun 2023

[CVPR 2023](#) | *Keywords: black-box attacks, adversarial robustness, dense prediction*

- Proposed a query-efficient black-box attack that aggregates surrogate gradients from model ensemble.
- Achieved high transferability and query-efficiency across detection and segmentation models.

### **Adversarial Robustness of Computational Imaging Systems**

Feb 2022 – Jul 2022

*Keywords: computational imaging, adversarial robustness, inverse problems*

- Evaluated the vulnerability of deep computational imaging models to adversarial perturbations.
- Compared learned imaging pipelines robustness against conventional non-learning methods and explored mitigation strategies.

## **ACADEMIC SERVICE**

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### **Conference Reviewer:**

- **2026:** ICLR, CVPR, ECCV, ICIP, WACV, NeurIPS, IEEE Asilomar.
- **2025:** ICCV, ICIP, IEEE Asilomar.
- **2024:** WACV.

### **Teaching Assistant:**

- UCR EE240 Pattern Recognition, Spring 2023 and Spring 2024.
- UCR CS171/EE142 Introduction to Machine Learning and Data Mining, Fall 2023 and Winter 2026.

## **TECHNICAL SKILLS**

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**Research Areas:** Trustworthy AI, safety steering, model-oversight, machine unlearning, adversarial robustness, safety optimization, vision-language models, text-to-image generation.

**Programming (ranked by proficiency):** Python, PyTorch, MATLAB, C/C++, HTML.

**Tools:** Git/GitHub, Linux/Unix, LaTeX, Shell, SQL, Qt.

**Communication:** 汉语/漢語 (Simplified/Traditional Chinese), English, Español (Spanish).